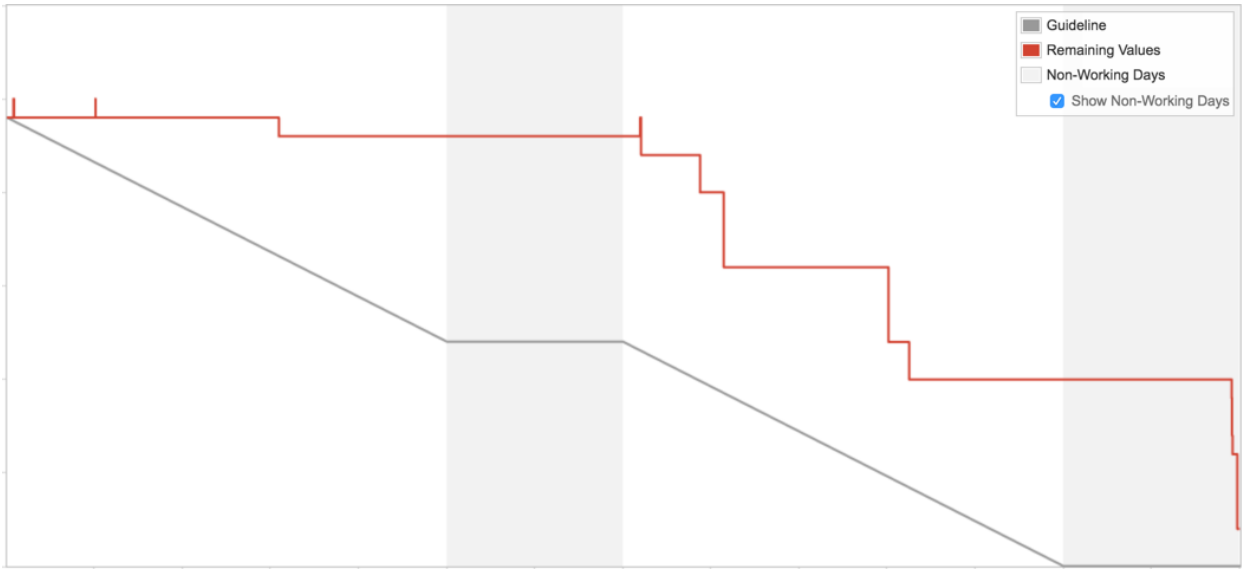
WEEKLY PROGRESS REPORT

General Information

Field	Value
Group Name	HealthCode
Week Number	2
Scrum Master	Sertaç Ataç
Date Submitted	17 March 2026
Domains Implemented	20 / 20

1. Burndown Chart



2. Completed This Week

US-004	Upload own CSV patient file	*2*
US-005	See columns with missing values	*3*

US-006	See patient and feature count summary	*4*
US-007	Choose how missing values are handled	*1*
US-008	Set train/test split ratio with slider	*3*
US-009	Choose feature scaling method	*2*
US-010	Change K value in KNN with live update	*4*
US-011	Select one of 6 ML algorithms	*5*
US-012	Adjust SVM hyperparameters	*3*
US-013	Visual comparison of algorithm characteristics	*2*
US-014	View confusion matrix after training	*5*
US-015	See accuracy, precision, recall, and F1-score	*3*
US-016	View ROC curve for trained model	*3*

3. In Progress

None — all sprint stories moved to **Done**.

4. Blocked / At Risk

No blockers identified this week.

5. Key Decisions Made

Z-score values computed from real data

Normalisation values were calculated directly from *cardiology.csv* ($\mu = 38.08$, $\sigma = 11.83$) instead of using estimates.

Minimum (14): -2.04
Maximum (80): 3.55

Schema gate with blocked state indicator

Implemented a strict validation gate requiring Column Mapper validation and saving before Step 3 unlocks.

A red **“Step 3 is Blocked”** banner appears if bypass is attempted.

Settings reset on modification

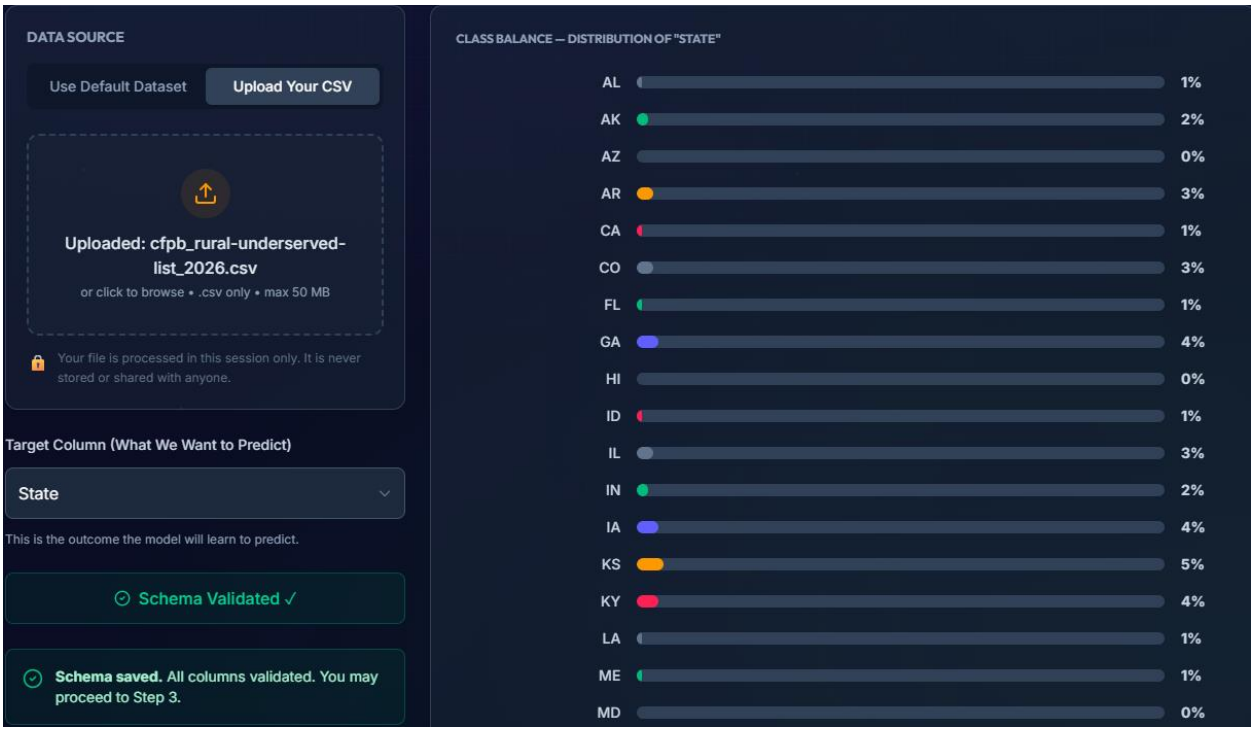
Any change to preparation settings resets the `isApplied` state, requiring re-application to maintain accurate visualisations.

6. Test Results

Story ID	Title	Acceptance Criteria Status
US-004	Upload own CSV patient file	☑ Pass
US-005	See columns with missing values	☑ Pass
US-006	See patient and feature count summary	☑ Pass
US-007	Choose how missing values are handled	☑ Pass
US-008	Set train/test split ratio with slider	☑ Pass
US-009	Choose feature scaling method	☑ Pass

Evidence:

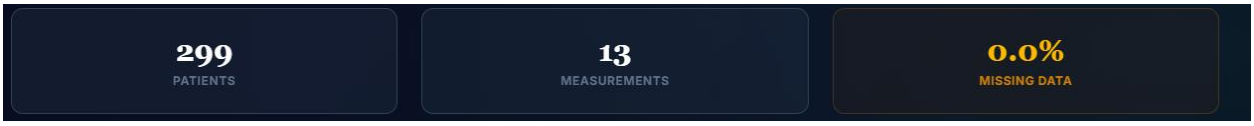
US-004



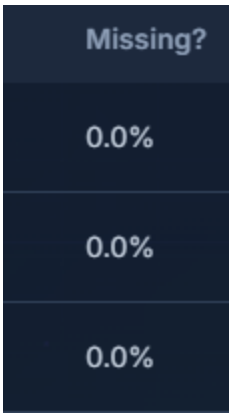
US-005

Measurement	Type	Missing?	Action Needed
age	Number	0.0%	Ready
anaemia	Number	0.0%	Ready
creatinine_phosphokinase	Number	0.0%	Ready

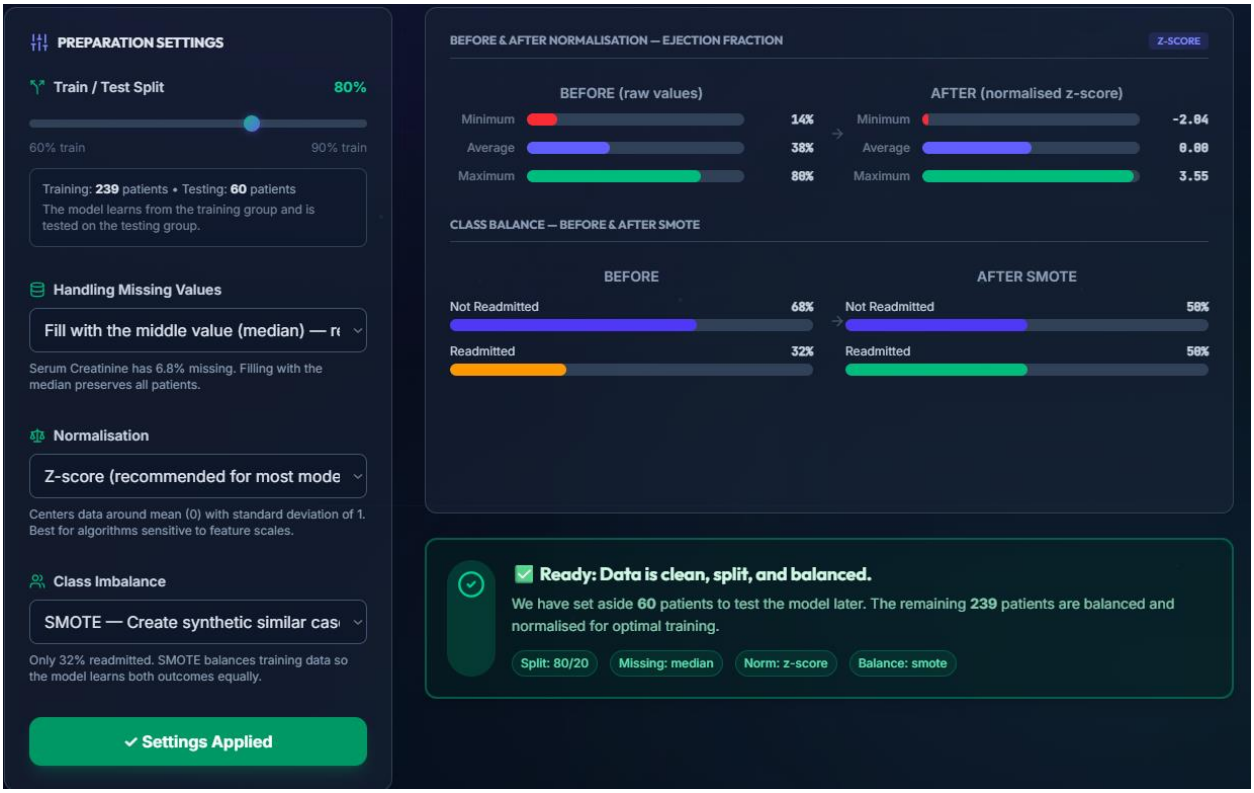
US-006



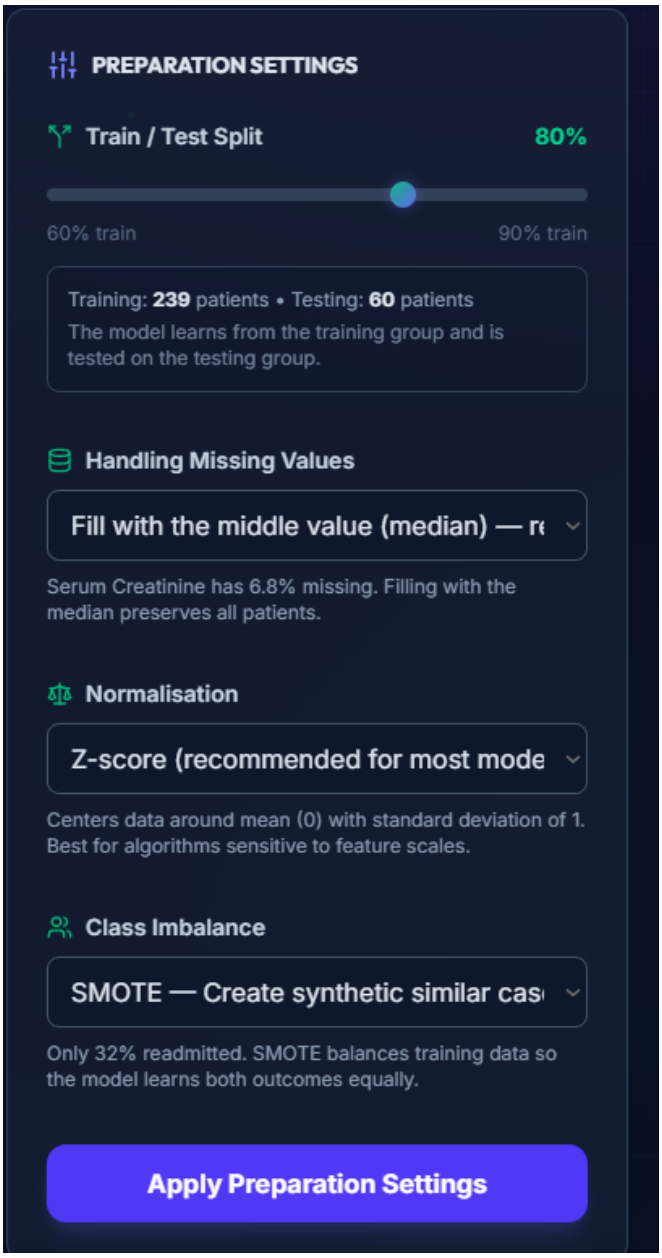
US-007


Missing?
0.0%
0.0%
0.0%

US-008


PREPARATION SETTINGS
Train / Test Split
80%
60% train
90% train
Training: 239 patients • Testing: 60 patients
The model learns from the training group and is tested on the testing group.
Handling Missing Values
Fill with the middle value (median) — r
Serum Creatinine has 6.8% missing. Filling with the median preserves all patients.
Normalisation
Z-score (recommended for most mode
Centers data around mean (0) with standard deviation of 1. Best for algorithms sensitive to feature scales.
Class Imbalance
SMOTE — Create synthetic similar cas
Only 32% readmitted. SMOTE balances training data so the model learns both outcomes equally.
✓ Settings Applied
BEFORE & AFTER NORMALISATION — EJECTION FRACTION
Z-SCORE
BEFORE (raw values)
Minimum 14%
Average 38%
Maximum 80%
AFTER (normalised z-score)
Minimum -2.04
Average 0.00
Maximum 3.55
CLASS BALANCE — BEFORE & AFTER SMOTE
BEFORE
Not Readmitted 68%
Readmitted 32%
AFTER SMOTE
Not Readmitted 50%
Readmitted 50%
Ready: Data is clean, split, and balanced.
We have set aside 60 patients to test the model later. The remaining 239 patients are balanced and normalised for optimal training.
Split: 80/20
Missing: median
Norm: z-score
Balance: smote

US-009



7. Metrics — Sprint 2

Metric	Target	Actual	Status
CSV Upload Success Rate	100%	100%	☑ Met
Column Mapper Gate	0 bypass bugs	0 bugs	☑ Met
Step 3 Controls	All functional	All functional	☑ Met
Domain Count (Step 1)	20 / 20	20 / 20	☑ Met

Test Coverage	100%	100% (18/18)	☑ Met

8. Next Week Plan — Sprint 3

Story ID	Title	Points	Assignee
US-4.1	Model selection dropdown (LR, DT, RF, SVM, kNN)	5	Emre Kurubaş
US-4.2	Hyperparameter controls per model	5	Emre Kurubaş
US-4.3	Training animation with progress indicator	3	Cem Özal
US-4.4	Model comparison table	3	Cem Özal
US-5.1	Confusion matrix visualisation	5	Sertaç Ataç
US-5.2	ROC curve chart	5	Can Eltayeb
US-5.3	Performance metrics (accuracy, precision, recall, F1)	3	Sertaç Ataç

9. Retrospective Note (Sprint 2)

Category	Note
Keep	Using real dataset statistics for visualisation ensured accuracy and helped catch errors early.
Improve	Begin writing test cases earlier in the sprint to avoid last-day rush and missed UI issues.
Try	Use pair programming for complex components like the Column Mapper to better manage multi-state flows.